Toward Better Accuracy-Efficiency Tradeoffs for Oriented SAR Ship Object Detection

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***Abstract*—In oriented synthetic aperture radar (SAR) ship detection task, convolutional neural network based detectors have dramatically improved the detection performance, but enormous parameters make it difficult to realize model lightweighting. Recently, DETR and its variants have demonstrated excellent performance in object detection task, while model construction through linear layers has great potential in terms of model lightweighting. However, DETR-based models are rarely applied to oriented object detection task, while the network structure relies on manual experience and cannot be designed automatically. In this article, we propose a novel neural architecture search based lightweight detector in polar coordinate system with DETR as search space for oriented SAR ship detection, where oriented bounding boxes are encoded and decoded in polar coordinate system to cope with boundary discontinuity problems, and the weight entanglement strategy is adopted to realize automatic and lightweight design of DETR. Meanwhile, we design an oriented multiscale attention to alleviate the problem of sampling a large amount of background due to offset learning. Furthermore, we introduce a downsampling feedforward network to significantly reduce network floating point operations. Finally, we transplant FPDDet head as auxiliary head to improve encoder potential ship feature learning and decoder cross-attention learning. Experimental results show that our models not only achieve DETR lightweighting and real-time detection, but also improve detection performance. Our base models achieve state-of-the-art performance on both RSSDD and RSDD datasets compared to previous best models, with 1.36% and 2.28% improvement in mAP with 32.67 and 32.14 GFLOPs, respectively.**

***Index Terms*—DETR, neural architecture search (NAS), oriented ship detection, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) image relies on the imaging characteristics of all-day, all-weather, and free of cloud interference, making it indispensable for ground-based maritime surveillance [1] and ship detection [2]. Therefore, accurate detection of SAR image ship objects is crucial. However, traditional SAR image ship detection methods, such as those relying on structural features based on ship contours [3], statistical properties of background clutter [4], and texture features based on grayscale information [5], struggle to achieve accurate detection of ship objects in complex background conditions and dense arrangement of inshore ships.

The rapid development of convolutional neural network (CNN) in object classification, instance segmentation, and detection tasks makes CNN in SAR ship classification [6], [7], instance segmentation [8], [9], [10], [11], and detection [12], [13], [14] tasks that have also been widely applied. At the same time, the evolution and optimization of the CNN framework, such as ShipDeNet-20 [15] and other lightweight methods [46], [47], can achieve more accurate detection with fewer parameters in tasks such as SAR ship lightweight detection. However, detection boxes of densely arranged inshore ships overlap heavily, making it difficult to screen detection boxes correctly. With the emergence of the oriented bounding box (OBB) based detection methods [16], [17], [18], which effectively solves the problems of overlapping detection boxes caused by horizontal bounding boxes (HBBs). However, the current OBB representation methods in the Cartesian coordinate system have the problems of boundary discontinuity, and the recently emerged methods in the polar coordinate system [19], [20] increase the inference time due to the complex encoding and decoding methods.

Although the OBB-based CNN methods can significantly solve the problems existing in HBB, the large number of parameters makes it difficult to lightweight oriented SAR ship object detectors. Recently, models such as ViT [21], swin transformer [22], DETR [23], and its variants [24], [25], [26] using transformer as the framework have shown outstanding performance in computer vision. However, HBB-based DETRs have several problems, such as the manual design of DETR model is time-consuming, as well as it is difficult to find the optimal model in a manually designed deep framework, HBB-based sample point offset learning samples large backgrounds in OBB-based detection tasks and how to select the auxiliary head when it is applied to oriented object detection tasks. For the aforementioned problems, we have provided our solutions accordingly in this article.

To automatically design the DETR architecture and implement real-time, lightweight detection of oriented SAR ship object detectors, neural architecture search (NAS) provides the solution to above problems. Recent weight-sharing-based NAS methods [27], [28], [29], [30] have been implemented to inherit the weights of the search space, thereby significantly reducing the search cost of subnets. Unlike the aforementioned CNN-based NAS methods, the weight entanglement method [31] is

Received 22 November 2024; revised 21 February 2025; accepted 23 March 2025. Date of publication 27 March 2025; date of current version 16 April 2025. This work was supported in part by the National Natural Science Foundation of China under Grant 62201114 and Grant 62301108; and in part by the Scientific Research Fund of Liaoning Provincial Education Department under Grant JYTMS20230171 and Grant JYTMS20230176. *(Corresponding authors: Buniu Niu; Si-nian Jin.)*

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Digital Object Identifier 10.1109/JSTARS.2025.3555330

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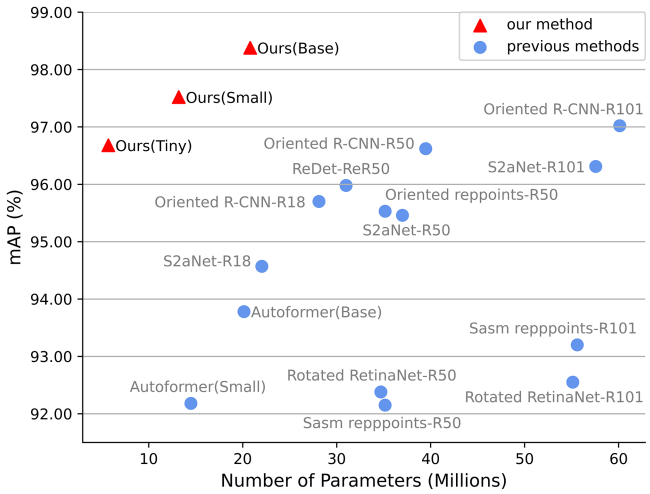


Fig. 1. Number of parameters (millions) vs. accuracy (mAP) on RSSDD. Our subnets based on the DETR search space with different model sizes are in comparison with the current mainstream oriented object detection models and NAS models, where R18, R50, ReR50, and R101 denote ResNet18, ResNet50, ReResNet50, and ResNet101, respectively. Note that all compared models are replicated for RSSDD. Our models achieve a tradeoff between the number of parameters and detection accuracy.

proposed to implement the pure transformer-based NAS model, where the transformer blocks share their common part weights.

In this article, we propose a novel NAS-based lightweight detector in polar coordinate system with DETR as search space for oriented SAR ship object detection. Specifically, we construct a DETR model in polar coordinate system to effectively reduce the number of parameters of oriented SAR ship detectors and alleviate the boundary discontinuity problem. Meanwhile, we introduce the weight entanglement strategy in NAS to automatically design the DETR framework and further lightweight the oriented ship detector. As shown in Fig. 1, our base model achieves state-of-the-art performance on RSSDD with only 20.8 M parameters, compared to the oriented R-CNN [32], our base model reduces the parameters by 65.37% and improves the AP value by 1.36%. Then, in the DETR encoder, we propose a downsampling feedforward network (FFN) to significantly reduce the floating point operations (FLOPs) of the overall network. Further, we propose an oriented multiscale attention in the DETR decoder to efficiently overcome the problem of HBB-based multiscale deformable attention (MSDA) [24] due to offset learning sampling a large amount of oriented ship object background. Furthermore, we transplant the FPDDet [33] detection head as the encoder auxiliary head to improve the encoder's learning of potential ship features, and unlike Co-DETR, we use the same OBB representation method throughout our network, which ensures the consistency of the regression of the whole network. In the decoder, we filter the positive samples of the FPDDet auxiliary head for the auxiliary training of the decoder to improve cross-attention learning. To validate the performance of our detector, we perform extensive experiments on RSSDD and RSDD datasets, and the experimental results confirm our NAS-based detector achieves better accuracy–efficiency tradeoffs for oriented SAR ship object detection.

Our contributions in this article can be summarized in the following five points.

1) We propose an NAS-based lightweight detector in polar coordinate system with DETR as search space for oriented SAR ship object detection, which cannot only effectively alleviate the problems of encoding and decoding complexity that exist in mainstream OBB representation methods, but also realize the lightweight design of DETR in polar coordinate system, and achieve the accurate and real-time detection of oriented ship objects.
2) We construct DETR detector based on the weight entanglement strategy in NAS to implement the automatic design of DETR and improve the FFN in DETR encoder, which effectively reduces model FLOPs.
3) We put forward an oriented multiscale attention for feature extraction of oriented ship objects in the DETR decoder, which effectively deals with the problem that HBB-based MSDA samples a large amount of background.
4) We introduce an auxiliary training method of the DETR encoder with FPDDet detection head to enhance potential ship feature learning and use the same polar encoding and decoding method to achieve consistency of network regression on OBB, while we further filter the positive samples in the auxiliary head for the auxiliary training of the decoder to improve cross-attention learning.
5) Our model achieves better accuracy–efficiency tradeoffs for oriented SAR ship object detection, while our base model achieves state-of-the-art performance, achieving 98.38% AP on RSSDD with 20.82 M parameters, 32.67 G FLOPs, and 30.25 ms average inference time, and 93.45% AP on RSDD with 25.78 M parameters, 32.14 G FLOPs, and 27.63 ms average inference time.

II . Related Works

A. SAR Image Ship Object Detection

SAR image ship object detectors are generally categorized into mainstream HBB-based detectors and recently emerged OBB-based detectors. Among the HBB-based SAR image ship object detectors, such as RetinaNet [34], Centernet [35], and Cornernet [36], some generalized HBB-based detectors are widely applied in SAR image ship object detection. Additionally, Wan et al. [37] employed YOLOX [38] as a lightweight design framework to reduce the complexity of the model. Feng et al. [39] introduced the positional attention module to enhance the features of ship objects.

In order to cope with the overlapping prediction boxes of HBB, various OBB-based methods have been proposed, among them, S2aNet [40] introduces the alignment convolution method to alleviate the feature mismatch between convolution and arbitrarily oriented objects. Oriented R-CNN [32] introduces an oriented region proposal network to improve the detection performance of the oriented object detector. For OBB-based SAR image ship detector, Sun et al. [41] achieved the prediction of the OBB of ship objects based on the mechanism that the ship objects appear to have strong scattering points in SAR images, which improves the detection efficiency of the ship object.

He et al. [20] achieved the encoding of the ship object of the SAR image by using a set of vectors under the polar coordinate system, which overcame the boundary discontinuity problem.

The above methods greatly mitigate the severe overlapping of prediction boxes caused by HBB. However, they still suffer from boundary discontinuity problem and encoding and decoding complexity.

B. SAR Image Ship Object Lightweight Detection

In order to guarantee the detection accuracy while achieving the real-time inference speed of the network, the lightweight methods for object detection networks have received great attention. In particular, MobileNet [42] splits the standard convolution into depthwise and pointwise convolution, which drastically reduces the number of parameters in the model and shortens the inference time. In addition, in transformer-based detectors, swin transformer [22] proposes the shifted window strategy to address the challenge of large variations in visual objects and to reduce the burden of high image resolution on hardware. DETR [23] suggests using query queue to eliminate postprocessing methods such as non-maximum suppression (NMS) to achieve end-to-end object detection. Co-DETR [26] proposes introducing multiple auxiliary heads to realize one-to-many positive samples learning of objects as a way to avoid the interference of encoder sparse supervision on feature learning. The above transformer-based frameworks reduce the number of detector parameters compared to CNN, which have great potential for the task of object detector lightweighting.

The CNN-based methods described above have been widely applied in lightweight detectors for SAR image ship objects, such as Zhou and Piao [43] used YOLOV4 as a framework to significantly reduce the network parameters of SAR image ship object detectors. Yang et al. [44] introduced split bidirectional feature pyramid network to reduce the interference from the background of SAR image. Li et al. [45] introduced feature jump connection structure and feature relay amplification network to realize lightweight detection. Zhang and Zhang [46] proposed a high-speed ship detection method in SAR images based on grid CNNs. This method improves detection accuracy and efficiency by adopting a grid-based strategy, which effectively handles complex backgrounds and variations in ship scale. Zhang et al. [47] proposed an efficient ship detection method for SAR images based on depth-separable CNNs. This method reduces computational complexity by using depth-separable convolutions, and improves detection accuracy and efficiency.

The mentioned lightweight methods show excellent detection performance in SAR image ship detection with a few network parameters, but they perform extensive and delicate manual design to reduce network parameters and accelerate network inference, which consumes a large amount of time.

C. Neural Architecture Search

NAS provides a solution to achieve lightweight and automatic design of neural network. Zoph et al. [48] proposed an architecture design paradigm that decouples the complexity of convolutional networks, which makes the design of convolutional architectures more efficient and versatile. DARTS [28] uses gradient descent to optimize neural network architectures in a continuous space, which avoids a huge expenditure of resources during the subnetwork search phase. PA&DA [49] improves the consistency and efficiency of one-shot NAS by optimizing the sampling distribution of paths and data based on the normalized gradient paradigm. AutoFormer [31] proposes a supernet training strategy based on weight entanglement, which makes it possible to search for pure transformer subnets. ViTAS [50] introduces a cyclic weight sharing strategy to eliminate the evaluation bias of subnetworks. ShiftNAS [51] achieved the purpose of allocating different computational resources for different parameter quantum subnets in the transformer architecture by adjusting the sampling probability to address the performance gap in the one-shot NAS.

For NAS-based SAR ship detectors, Li et al. [52] proposed a DARTS based method to realize the automatic design of SAR ship detector. SarNAS [53] introduces a hardware-aware SAR object detection algorithm that achieves efficient performance optimization through multiobjective NAS.

The above methods provide rich means for constructing search spaces and subnet search strategies. However, there is still a lack of exploration in constructing DETR-based search space and implementing the automatic design of oriented ship object detector.

III. Proposed Method

This section first introduces the overall structure of the NAS-based DETR lightweight detector in Section III-A, followed by the DETR search space implemented based on the weight entanglement strategy in Section III-B. Then, the oriented multiscale attention is described in Section III-C. We will then show the auxiliary head of the DETR encoder and the auxiliary training method of the decoder in Section III-D. Finally, in Section III-E, we describe the loss function for the entire search space.

A. Pipeline

Our NAS-based lightweight detector uses DETR as the framework and represents OBB in polar coordinate system for oriented ship detection, which overcomes the boundary discontinuity problem. The search space is built based on the weight entanglement strategy, which further makes the network lightweight and realizes the automatic design of the DETR framework. In the FFN of the DETR encoder, we simply downsample part of the multiscale features to dramatically reduce the FLOPs of whole network. Unlike deformable DETR [24], which uses MSDA in the decoder to select sampling points based on HBB, we propose oriented multiscale attention based on OBB for sampling to realize the fusion and extraction of multiscale features of oriented ship objects. Inspired by Co-DETR, we transplant the detection head of FPDDet as the encoder auxiliary head to improve the encoder's learning for potential ship features, and unlike Co-DETR, the auxiliary head and the decoder are aligned in our OBB regression method, which ensures the consistency of the whole network regression, and we further filter the positive samples of the auxiliary head for the auxiliary training of the decoder to improve cross-attention learning. As Fig. 2 shows, the NAS-based lightweight DETR detector consists of three parts:

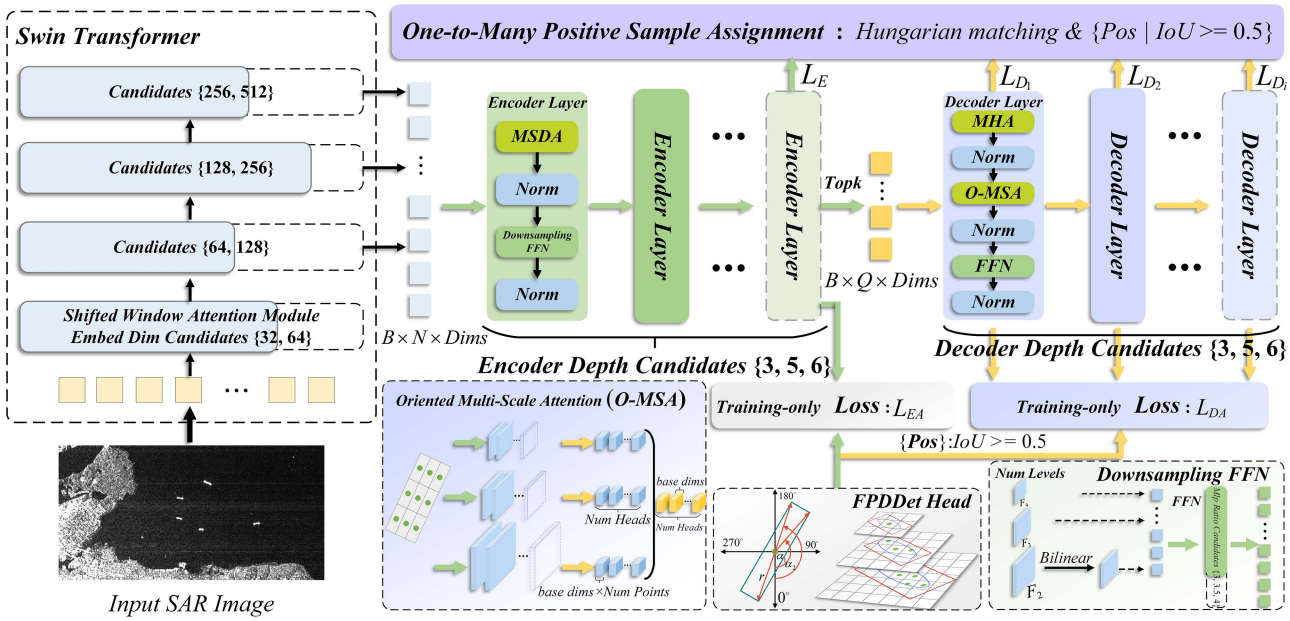


Fig. 2. Architecture of NAS-based lightweight detector in polar coordinate system with DETR as search space for oriented SAR ship detection. Our NAS model includes backbone based on swin transformer, encoder and decoder based on DETR. Our NAS model based on DETR search space is implemented based on the weight entanglement strategy, where the dimension of each stage in the backbone and the number of layers in the encoder and decoder is taken as candidates. In the encoder, MSDA denotes multiscale deformable attention, downsampling FFN is proposed to reduce the FLOPS of the network, and in the decoder, MHA denotes multihead attention, oriented multiscale attention (O-MSA) is proposed to extract the ship object features. In the training phase, FPDDet head is introduced for auxiliary training of the encoder, while filtered samples from FPDDet head are used for auxiliary training of the decoder. The encoder and decoder are trained using one-to-many positive sample assignment strategy. In the inference phase, the auxiliary training part is discarded.

backbone, encoder, and decoder, which is described in detail below.

The backbone network is constructed using swin transformer. To automatic search the swin transformer framework, we set the embedding dimensions, the number of heads in the multihead attention, the MLP ratio of each stage in the swin transformer, and the embedding dimensions of the backbone output features as dynamic candidates. The input SAR image channels are first adjusted and the image resolution is downsampled to a quarter by dynamic patch embedding operation. In the following shifted window attention modules [22], the output dimensions of each stage are different, which can significantly reduce the FLOPs of the network compared to attention modules in AutoFormer that output the same embedding dimensions, under the condition of inputting SAR images of the same size. The top three layers features $\{F_2, F_3, F_4\}$ of swin transformer are adjusted embedding dimensions by linear layers and used as the output of the backbone and as inputs to the DETR encoder directly.

In the encoder, we construct the transformer framework, which can automatically design based on the MSDA module, and make the number of encoder layers as the dynamic candidate of NAS to further reduce the cost of manually designing the network, the encoder outputs the class score $P_{ES} \in [0,1]^{B\times N\times 1}$ and the predicted OBB $P_{EO} \in \mathbb{R}^{B\times N\times 5}$, where B is the batch size, N is the number of features, and the output channel of P_{EO} is determined by the polar encoding method of FPDDet. Then, considering the input of multiscale features will cause the FLOPs of FFN in the encoder to increase dramatically. In order to effectively reduce the FLOPs, we downsample part of the input feature layer to realize the reduction of FLOPs. Finally, we transplant the detection head of FPDDet as an auxiliary head to auxiliary train the encoder to improve the encoder's learning for potential ship features.

For the decoder, we also take the number of decoder layers as a dynamic candidate for the NAS. Meanwhile, to realize the precise feature extraction of oriented ships, unlike the HBB-based MSDA module, we propose oriented multiscale attention to realize the fusion and feature extraction of oriented ships. To improve the cross-attention learning of the decoder, we further filter the prediction OBB of the FPDDet auxiliary head for the auxiliary training of the decoder. It is worth noting that the auxiliary head and the auxiliary training of the decoder can be discarded in the inference stage. The outputs of each decoder layer consist of $P_{DS} \in [0,1]^{B\times Q\times 1}$ and $P_{DO} \in \mathbb{R}^{B\times Q\times 5}$, where Q denotes the number of queries.

B. Search Space

We construct a DETR search space based on the weight entanglement strategy in polar coordinate system to realize lightweight, real-time detection of oriented ship objects and automatic design of our detector, where the dynamic candidates include embedding dimensions of each stage in the backbone, embedding dimensions in encoder and decoder, number of heads, MLP ratio (the ratio of hidden dimension to the embedding dimension in the multilayer perceptron), encoder depth, and decoder depth. As shown in the details of Table I, unlike AutoFormer which stacks multiple layers of dynamic

TABLE I
DETR SEARCH SPACE BASED ON WEIGHTED ENTANGLEMENT STRATEGY

	Backbone	Encoder	Decoder
Embed dim	/	(128 ,192, 256)	(128 ,192, 256)
Head num	(6, 7, 8)	(6, 7, 8)	(6, 7, 8)
MLP ratio	(3.0, 3.5, 4.0)	(3.0, 3.5, 4.0)	(3.0, 3.5, 4.0)
Encoder depth	/	(3, 5, 6)	/
Decoder depth	/	/	(3, 5, 6)
Embed dim in backbone	[(32, 64), (64, 128), (128, 256), (256, 512)]		
Params range	4.2–23.6M		

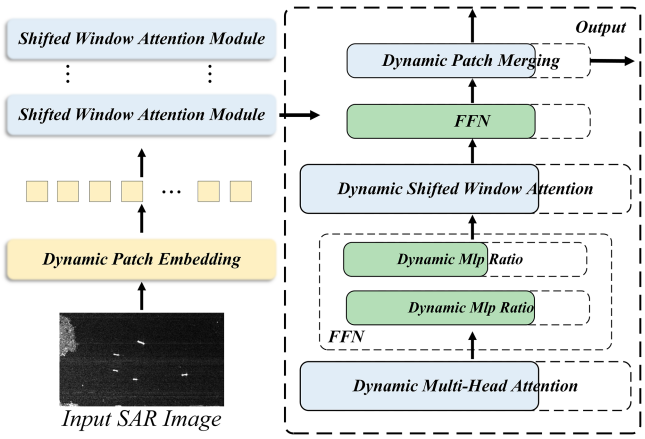


Fig. 3. Swin transformer backbone network based on weighted entanglement strategy.

multihead attention, we construct the search space of oriented ship detector based on the weight entanglement strategy for the backbone, encoder, and decoder of DETR, respectively. To further reduce the cost of manually designing neural networks, we set the embedding dimensions in each stage of the backbone and the encoder and decoder depths of DETR as dynamic candidates.

As depicted in Fig. 3, the backbone structure is based on the swin transformer. The input SAR image is first adjusted for channel and resolution by a 4×4 convolution (conv) in the dynamic patch embedding operation, and then fed into the shifted window attention module, which contains a dynamic multihead attention and a dynamic shifted window attention. Following each attention, an FFN is constructed, while following each layer of the attention module, a dynamic patch merging operation is constructed to downsample the output feature. The backbone is composed of four layers of the abovementioned module and takes the top three layers as outputs of the backbone, whose feature sequence lengths are $H/8 \times W/8$, $H/16 \times W/16$, and $H/32 \times W/32$, respectively, where H and W denote the height and width of the input SAR image. As shown in Table I, different from the AutoFormer, the embed dimensions of each stage in our backbone are different, which effectively reduces the FLOPs of our network and further implements the automatic design of the backbone in the oriented SAR ship detector.

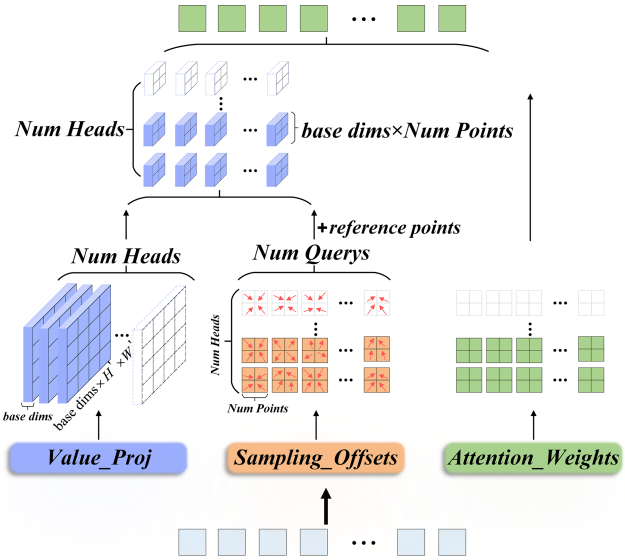


Fig. 4. Architecture of multiscale deformable attention based on weight entanglement strategy in encoder.

In the encoder, we take the number of MSDA layers as a dynamic candidate, while the linear layers in the encoder are constructed based on the weight entanglement strategy in NAS. Fig. 4 illustrates that the features are initially processed by the value linear layer to adjust the channel dimensions of features according to the number of dynamic heads. The features are processed by the sampling linear layer to generate the sampling point offset of queries, which are then used to sample feature values at the corresponding positions in the feature map. The weight linear layer generates weights that are used to weight and sum the sampled values, resulting in the output of the MSDA. In addition, each encoder layer contains a dynamic FFN. As shown in Fig. 2, to effectively reduce the FLOPs of the FFN, we bilinearly downsample the F_2 feature layer in the multiscale features and then upsample it after the linear layer and sum it with the input queries as the output of the FFN. Ultimately, the output of the dynamic FFN is passed through the layernorm as the output of the encoder layer.

The decoder in our NAS sets the number of layers to dynamic candidate. However, the decoder differs from the encoder in that it selects a fixed number of features as the query queue compared to multiscale features, which significantly reduces the network FLOPs and lowers the network inference time. First, the decoder uses the dynamic multihead attention to establish relationships between objects and enhance the representational ability of the model. As shown in Fig. 5, the feature sequences are fed into the value linear layer to generate the feature map. The sampling point is extracted by decoding OBBs at the corresponding position of the queries in the encoder output, and in contrast to the MSDA mechanism employed in the encoder, there is no offset learning at the sampling points, which not only reduces the number of network parameters but also ensures the accuracy of ship object detection. The feature values are then sampled at the corresponding positions of the feature map. The weights are

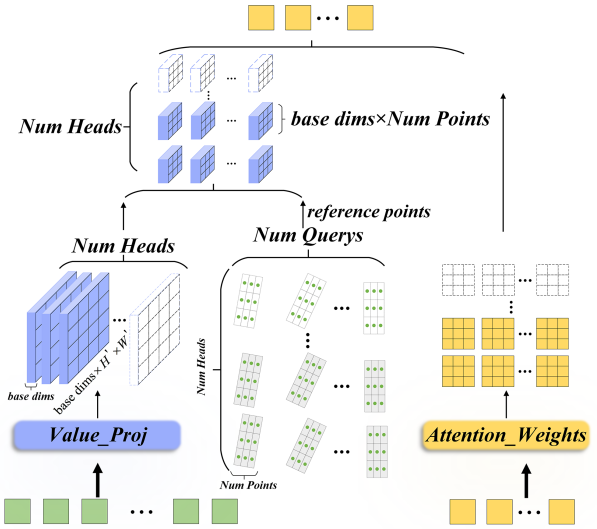


Fig. 5. Architecture of oriented multiscale attention based on weight entanglement strategy in decoder.

generated by the weight linear layer. Then, the feature values are weighted and summed to obtain the outputs of the oriented multiscale attention. Finally, output feature sequences of the decoder layer is obtained through the dynamic FFN.

C. Oriented Multiscale Attention

DETR [23] and its variants [24], [25], [26] based on MSDA use four point offsets of the HBB center point about width and height as sampling points in the decoder to sample feature values from the feature map, which shows excellent performance in the HBB-based object detection task, but when it is applied to the oriented object detection task, especially the oriented ship object detection task that contains long and slender ships, as illustrated in Fig. 6(b), due to the characteristics of the long and slender shape of the SAR image ship objects, there is the problem that the sampling points sample a large amount of SAR image background. In order to solve the above problem, inspired by Rotated ROI Align [16], we propose oriented multiscale attention as a solution to sample the grid points of the OBB directly, thus avoiding the problem of sampling the background, and at the same time eliminates the learning of sampling point offsets, which reduces the parameters of the DETR, and further realizes a lightweight design of the detector. Algorithm 1 shows the oriented multiscale attention method based on weight entanglement strategy.

In Algorithm 1, the weights and biases of linear layers in the oriented multiscale attention method are first obtained based on the weight entanglement strategy for implementing automatic search and lightweight design of NAS, which includes weight_v, bias_v for mapping the encoder output, weight_a, bias_a for weighted sum of sampled points, and weight_o, bias_o for mapping oriented multiscale attention output.

In contrast to HBB-based MSDA, as shown in Fig. 6(c), we do not learn the offsets for the sampling points, but directly

Algorithm 1: Oriented Multiscale Attention Based on Weight Entanglement Strategy.

input : $\boldsymbol{q}$: Queries of DETR (num_query,bs,dynamic_embed_dims).
$\boldsymbol{v}$: Encoder output (num_key,bs,dynamic_embed_dims).
$\boldsymbol{query_pos}$: The positional encoding for query (num_query,bs,dynamic_embed_dims).
$\boldsymbol{reference_points}$: Encoder predicted OBBs based on polar encoding filtered by decoder (bs,num_query,num_levels,5).
$\boldsymbol{dynamic_embed_dims}$: dynamic embed dims in weight entanglement strategies.
$\boldsymbol{dynamic_num_heads}$: dynamic num heads in weight entanglement strategies.
$\boldsymbol{factors}$: $[img_w, img_h, img_w, 10\pi, 10\pi]$.
output: output fused SAR image multi-scale features (num_query,bs,dynamic_embed_dims).

Obtain value_proj linear layer weight and bias: $\boldsymbol{weight_v}, \boldsymbol{bias_v}$, attention_weights linear layer weight and bias: $\boldsymbol{weight_a}, \boldsymbol{bias_a}$, out_proj linear layer weight and bias: $\boldsymbol{weight_o}, \boldsymbol{bias_o}$ based on input dynamic_embed_dims and dynamic_num_heads;

The $\boldsymbol{reference\ points}$ are multiplied by $\boldsymbol{factors}$ to decode the center point$(\boldsymbol{x_r}, \boldsymbol{y_r})$, polar diameter$(\boldsymbol{r})$, and polar angles$(\boldsymbol{\alpha_1}, \boldsymbol{\alpha_2})$ of OBB: $(\boldsymbol{x_r}, \boldsymbol{y_r}, \boldsymbol{r}, \boldsymbol{\alpha_1}, \boldsymbol{\alpha_2}) \leftarrow \boldsymbol{reference_points} \times \boldsymbol{factors}$;

Obtain four corner points of OBB: $\begin{bmatrix} x'_1 & x'_2 \\ y'_1 & y'_2 \end{bmatrix} \leftarrow \boldsymbol{r} \begin{bmatrix} \cos(\frac{\pi}{2} - \alpha_1) & \cos(\frac{\pi}{2} - \alpha_2) \\ \sin(\frac{\pi}{2} - \alpha_1) & \sin(\frac{\pi}{2} - \alpha_2) \end{bmatrix} + \begin{bmatrix} x_r & x_r \\ y_r & y_r \end{bmatrix}$; obtain the remaining two corner points $\begin{bmatrix} x'_3 & x'_4 \\ y'_3 & y'_4 \end{bmatrix}$ based on symmetry and further obtain grid points $\begin{bmatrix} x_1 & \dots & x_8 \\ y_1 & \dots & y_8 \end{bmatrix}$;

$\boldsymbol{sampling\ locations} \leftarrow \begin{bmatrix} x_r/img_w & \dots & x_8/img_w \\ y_r/img_h & \dots & y_8/img_h \end{bmatrix}$;

$\boldsymbol{value} \leftarrow$ Linear($\boldsymbol{v}$,$\boldsymbol{weight_v}$,$\boldsymbol{bias_v}$);

$\boldsymbol{attention_weights} \leftarrow$ Linear($\boldsymbol{q}$+$\boldsymbol{query_pos}$,$\boldsymbol{weight_a}$,$\boldsymbol{bias_a}$);

Reshape $\boldsymbol{value}$ to feature map $\boldsymbol{v'} \leftarrow (v'_1, v'_2, \dots v'_{num_levels})$;

for v'_i *in* $\boldsymbol{v'}$ **do**
    Sampling values $\boldsymbol{output_i}$ are obtained from $\boldsymbol{sampling\ locations}$ of v'_i;
end

$\boldsymbol{output_{tmp}} \leftarrow$ Weighted sum of ($\boldsymbol{output_i}$,... $\boldsymbol{output_{num_levles}}$) and $\boldsymbol{attention_weights}$;

Return $\boldsymbol{output} \leftarrow$ $\boldsymbol{q}$+Linear($\boldsymbol{output_{tmp}}$,$\boldsymbol{weight_o}$,$\boldsymbol{bias_o}$).

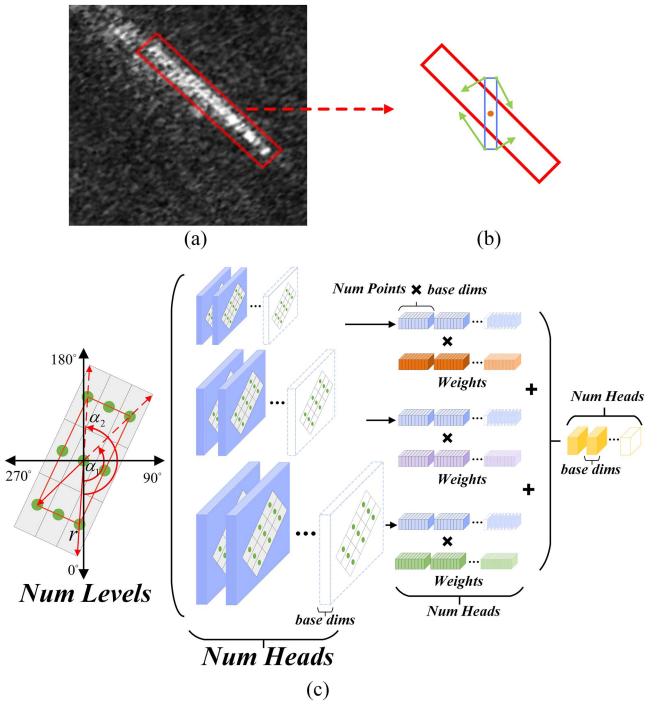


Fig. 6. (a) Input SAR ship image. (b) Sampling point offset learning method for HBB-based multiscale deformable attention, where green arrows indicate HBB-based offset learning. (c) Oriented multiscale attention, where $r$indicates the polar diameter from the center of the OBB to the corner point, α_1 and α_2 indicate the two polar angles, and the green dots indicate the locations of the sampling points.

sample the grid points of the OBB. This is mainly due to the long and slender characteristic of the SAR image ship objects and the problem of sampling a large amount of background with the sampling points based on offset learning. Therefore, we then decode the OBBs filtered from the DETR encoder to obtain grid points of OBBs as sampling points, where the OBB is encoded as shown in Fig. 2, based on the five parameters in FPDDet [33] to effectively address the problems of boundary discontinuity and complexity of encoding and decoding. As illustrated in Algorithm 1, the grid points of OBB can be obtained by the following steps: first, the reference points filtered from the encoder are multiplied by factors to obtain $(x_r, y_r, r, \alpha_1, \alpha_2)$, further, as shown in Fig. 6(c), the four corner points can be computed through

$$\begin{bmatrix} x'_1 & x'_2 \\ y'_1 & y'_2 \end{bmatrix} = r \begin{bmatrix} \cos(\pi/2 - \alpha_1) & \cos(\pi/2 - \alpha_2) \\ \sin(\pi/2 - \alpha_1) & \sin(\pi/2 - \alpha_2) \end{bmatrix} + \begin{bmatrix} x_r & x_r \\ y_r & y_r \end{bmatrix} \tag{1}$$

and the remaining two corner points are obtained by symmetry, while further calculations are performed to obtain the grid points $\begin{bmatrix} x_1 & \dots & x_8 \\ y_1 & \dots & y_8 \end{bmatrix}$. Finally, the grid points are obtained as the sampling locations of the feature map by

$$\text{samplinglocations} = \begin{bmatrix} x_r/\text{img_}w & \dots & x_8/\text{img_}w \\ y_r/\text{img_}h & \dots & y_8/\text{img_}h \end{bmatrix}. \tag{2}$$

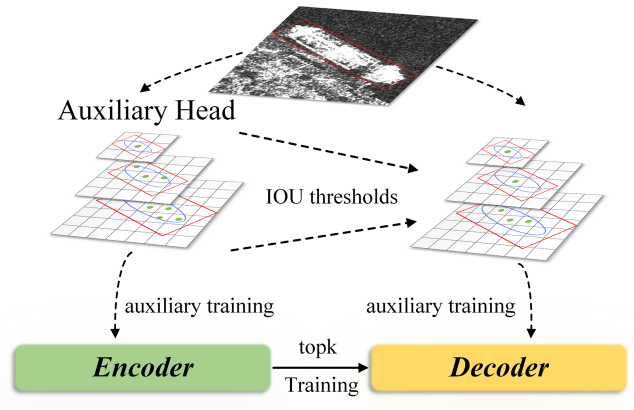


Fig. 7. Auxiliary training method for DETR detector based on oriented SAR ship objects, in which FPDDet head is used for auxiliary training of encoder to enhance the learning of potential ship object features, and further filtering of positive samples from FPDDet head realizes auxiliary training of decoder to improve cross-attention learning. **Note:** The auxiliary training part is discarded in the inference phase.

In the multiscale feature fusion stage, the encoder output is processed as follows:

$$\text{value} = \text{Linear}(v, \text{weight}_v, \text{bias}_v) \tag{3}$$

where v denotes the encoder output. For the weight of each sample point in the multiscale feature, we have

$$\text{attention_weights} = \text{Linear}(q + \text{query_pos}, \text{weight}_a, \text{bias}_a) \tag{4}$$

where q denotes the queries of DETR decoder and query_pos denotes the positional encoding of the queries. As shown in Fig. 6(c), the features corresponding to the sample points can be extracted and weighted and summed with the attention_weights to obtain $\text{output}_{\text{tmp}}$, and the output of oriented multiscale attention can be obtained by

$$\text{output} = q + \text{Linear}(\text{output}_{\text{tmp}}, \text{weight}_o, \text{bias}_o). \tag{5}$$

D. Encoder Auxiliary Head and Decoder Auxiliary Training Method

Co-DETR, a variant of DETR, enhances potential feature learning by introducing auxiliary heads at the encoder and improves cross-attention learning by assigning one-to-many positive samples to objects in the decoder for auxiliary training. Inspired by the above ideas, and in order to apply to the task of oriented ship object detection, as shown in Fig. 7, we transported the FPDDet detection head as the auxiliary head of the encoder, which uses dense regression strategy to realize one-to-many assignment of positive samples, to enhance potential features learning of ship objects. Meanwhile, we further filter out the OBB predicted by the auxiliary head to improve the cross-attention learning of our decoder.

Different from Co-DETR, our ship's OBB is regressed using the consistent polar encoding method throughout the network, as shown in Fig. 2, which ensures the consistency of the network's regression on the OBB. As shown in Fig. 7, we transport the FPDDet detection head for auxiliary training of the encoder, and

in the decoder, in addition to selecting a fixed number of queries (the number of queries is set to 300) based on the encoder's predicted scores for the decoder's forward training, we further filter the predicted OBB of the auxiliary head according to the evaluation metrics (i.e., IoU thresholds greater than or equal to 0.5) from the dense regression positive samples and use them for auxiliary training of the decoder.

E. Loss Function

The losses of the whole network include four parts, as shown in Fig. 2, corresponding to the encoder, decoder, FPDDet auxiliary head, and decoder auxiliary training parts in our lightweight DETR detector, each part includes losses on the predicted scores and OBB. The difference with DETR is that in the encoder and decoder score loss functions, in addition to the one-to-one Hungarian matching between the ship's target and the predicted boxes, the predicted boxes of the corresponding positions where the IoU between the regression's OBB and the ground truth is greater than or equal to 0.5 are set to positive samples. In the experimental section, we further compare the effect of the one-to-one Hungarian matching method with the above positive sample assignment method on the detection of oriented ship objects in SAR images.

For the score loss L_{ES} in the encoder, we use Quality Focal Loss [54] to calculate the loss between the predicted scores and the target scores. The target scores with IoU greater than or equal to 0.5 are first set to their IoU values, and then the target scores at the corresponding positions are set to 1 according to the Hungarian matching results. The loss between predicted scores and target scores can be calculated as follows:

$$L_{\text{ES}} = \frac{1}{N_S}\sum_{i=0}^{N_{\text{query}}} \times \begin{cases} L_{\text{BCE}}(P_{\text{ES}_i}, T_{\text{ES}_i})|T_{ES_i} - P_{\text{ES}_i}|^{\beta} W_{\text{ES}_i} & W_{\text{ES}_i} = 1 \\ L_{\text{BCE}}(1 - W_{\text{ES}_i}) & \text{otherwise} \end{cases} \tag{6}$$

$$L_{\text{BCE}} = -(T_{\text{ES}_i})\log(P_{\text{ES}_i}) - (1 - T_{\text{ES}_i})\log(1 - P_{\text{ES}_i}) \tag{7}$$

where N_S denotes the number of positive samples, N_{query} denotes the number of queries, $P_{\text{ES}} \in [0, 1]$ denotes the predicted scores of the encoder, $T_{\text{ES}} \in [0, 1]$ denotes the target scores, W_{ES} denotes the sample weight (1 for positive samples and 0 for negative samples), and β is a hyperparameter of Quality Focal Loss, which is set to 2 in the experiments.

For the regression OBB loss L_{EO} in the encoder, we use the L1 loss to calculate the loss between the predicted OBB and the ground truth, and it is implemented as follows:

$$L_{\text{EO}} = \frac{1}{N}\sum_{i=0}^{N_{\text{query}}} |P_{\text{EO}_i} - T_{\text{EO}_i}| W_{\text{EO}_i} W_{\text{EO}_i} = 1 \tag{8}$$

where N denotes the number of positive samples, P_{EO} denotes the predicted OBB, T_{EO} denotes the OBB target, and W_{EO} denotes the sample weight (1 for positive samples and 0 for negative samples).

For the whole loss of the encoder, we have

$$L_E = L_{\text{ES}} + \lambda_1 L_{\text{EO}}. \tag{9}$$

In the auxiliary head, for the predicted feature map and the generated Gaussian heatmap, we use Gaussian Focal Loss [36] to compute the loss between them, which is calculated by

$$L_{\text{EAS}} = \frac{1}{N_A}\sum_{i=0}^{N_{\text{levels}}}\sum_{j=0}^{H'-1}\sum_{k=0}^{W'-1} L_{\text{GFL}}(P_{A_{ijk}}, T_{A_{ijk}}) \tag{10}$$

$$L_{\text{GFL}} = \begin{cases} -(1 - P_{A_{ijk}})^{\alpha}\log(P_{A_{ijk}})W_{A_{ijk}} & W_{A_{ijk}} = 1 \\ -(P_{A_{ijk}})^{\alpha}\log(1 - P_{A_{ijk}})(1 - T_{A_{ijk}})^{\gamma} & \\ (1 - W_{A_{ijk}}) & \text{otherwise} \end{cases} \tag{11}$$

where L_{GFL} is the Gaussian Focal Loss,N_A is the number of dense regression positive samples, N_{levels} denotes the feature map level numbers, H' and W' denote the feature map height and width, P_A denotes the prediction heatmap, T_A denotes the ship Gaussian heatmap, W_A is the positive sample weights, and α and γ are the two hyperparameters in the Gaussian Focal Loss, which are 2 and 4, respectively.

The total loss of the auxiliary head can be calculated as follows:

$$L_{\text{EA}} = L_{\text{EAS}} + \lambda_1 L_{\text{EAO}} \tag{12}$$

where L_{EAO} is the L1 loss of OBB in the auxiliary head.

In the decoder loss, different from the encoder, each layer contains a Quality Focal Loss and an L1 Loss, which can be calculated as follows:

$$L_D = \sum_{i=0}^{N_{\text{layers}}} (L_{\text{DS}_i} + \lambda_1 L_{\text{DO}_i}) \tag{13}$$

where N_{layers} denotes the number of layers in the decoder,L_{DS} denotes the Quality Focal Loss in the decoder with a positive sample assignment method consistent with that in the encoder, and L_{DO} denotes the L1 loss of the OBB in the decoder.

For the loss in decoder auxiliary training, we also have a Quality Focal Loss and an L1 loss in each layer, and for the target score in Quality Focal Loss, the score is the IoU value between the predicted OBB and the ground truth. For the total loss in decoder auxiliary training, we have

$$L_{\text{DA}} = \sum_{i=0}^{N_{\text{layers}}} (L_{\text{DAS}_i} + \lambda_1 L_{\text{DAO}_i}). \tag{14}$$

The total loss of the DETR-based oriented ship target detector is given by the following equation:

$$\text{Loss} = L_E + L_{\text{EA}} + L_D + L_{\text{DA}}. \tag{15}$$

IV. Experiments

In order to verify the performance of the proposed NAS-based lightweight detector with DETR as the search space for oriented SAR ship object detection and that it outperforms the current state-of-the-art oriented SAR ship object detectors, we perform

TABLE II
COMPARATIVE RESULTS OF ORIENTED SHIP OBJECT DETECTION IN THE RSSDD DATASET UNDER THE CNN, TRANSFORMER, AND DETR NETWORKS

Models	Parameters (M)	FLOPs (G)	Resolution	Inshore AP.5	Offshore AP.5	All AP.5
DARTS- tiny [28]	13.21	26.99	608^2	35.01	72.07	60.09
Autoformer- tiny [31]	6.15	35.44		75.01	96.90	90.77
Ours- tiny	**5.78**	**10.84**		**89.30**	**99.23**	**96.68**
DARTS- small [28]	26.99	44.83	608^2	39.70	73.23	62.62
Autoformer- small [31]	14.49	83.54		76.02	98.34	92.18
Ours- small	**13.25**	**16.64**		**92.64**	**99.48**	**97.52**

evaluation and comparison experiments with the current mainstream oriented object detectors and lightweight detectors on RSSDD [55] and RSDD [56], two public oriented SAR ship datasets.

A. Implementation Details

In the training and testing phases of the lightweight DETR detector, all experiments are performed on an RTX 4090 GPU. For the training of the DETR search space, we use the adaptive moment estimation with decoupled weight decay (AdamW) optimizer to train 500 epochs, where the learning rate is initialized to 1×10^{-4}, the weight decay is set to 1×10^{-4}, and CosineAnnealingLR is adopted as the learning rate strategy. For the subnet, we use the same strategy as in the training phase and perform 150 epochs of fine tuning. We set the number of learnable object queries to 300 and set $\{\lambda_1\}$ to $\{5\}$ by default. During training and testing, the image resolution of RSSDD is resized to 608×608 and the image resolution of RSDD is kept at 512×512.

B. Ablation Study

The RSSDD dataset is used for all ablation experiments with the searched DETR tiny model as baseline, where the VOC2012 [57] AP.5 metric is used to evaluate the detection performance.

1) DETR Search Space Based on Weighted Entanglement Strategy: To demonstrate the advantages of our DETR-based subnet over those based on CNN or transformer, we conducted comparative experiments of subnets under different frameworks in terms of number of parameters, FLOPs, average inference time, and detection performance. As illustrated in Table II, our tiny model has been found to reduce the number of parameters by 56.25%, and the number of FLOPs by 59.84 G, while significantly improving the detection performance in comparison to the DARTS-tiny model. This is due to the fact that our model is constructed using linear layers, with the number of parameters in the convolutional layers being significantly reduced. Compared with the DARTS-small model, the detection performance of our-small model is improved by 34.90%, while the number of parameters is reduced by 50.91% and the FLOPs is reduced by 28.29 G. Compared with the searched Autoformer-tiny model based on transformer, with a similar number of parameters, our tiny model significantly reduces the FLOPs by 24.60G, and improves the performance by 5.91%. The considerable reduction in FLOPs can be attributed to three principal factors. First, the backbone of our model is constructed based on swin transformer, which allows for a gradual increase in the embedding dimension. This is in contrast to the Autoformer model. Second, the downsampling of part of the feature layer in the encoder also contributes to a reduction in FLOPs. The most significant factor is DETR decoder employs a fixed number of queries to perform the detection tasks. The fixed number of queries effectively reduces the FLOPs of the network in comparison to inputting multiscale features. The presented experiments demonstrate that our model not only significantly reduces the number of network parameters in terms of model lightweighting, but also reduces the network's FLOPs and significantly improves the detection performance of oriented ship objects in SAR images.

2) Improved Downsampling FFN in Encoder: Taking the supernet with an input SAR image resolution of 608×608 as an illustrative example, before the FFN in the encoder is not improved, the FLOPs of the supernet are 54.64G, of which the backbone is 15.17G, accounting for 27.76%, one layer of the encoder is 5.54G, the whole encoder is 33. 24G, accounting for the supernet's 60.87%, of which the FLOPs of FFN in an encoder layer is 3.97G, and the FFN in the encoder accounts for 43.59% of the supernet's FLOPs, the FLOPs of a decoder layer are 0.92G, and the FLOPs of the decoder accounts for only 10.10% of the supernet's FLOPs. The FLOPs of FFN in the improved encoder layer are only 1.70G, which is 57.18% less compared to the FLOPs of the unimproved FFN, and the FLOPs of the supernet are 40.96G, which is 25.04% less FLOPs compared to the previous supernet. To further compare the effects of FFN before and after improvement on the performance of oriented ship object detection in SAR images, we sampled 1000 subnets of the supernet to compare the AP values and subnet FLOPs. As shown in Fig. 8(a) and (b), in the case of similar number of parameters of subnets, the FLOPs of FFN after improvement are significantly reduced, and at the same time, the performance of oriented ship object detection is less affected. In the case of similar FLOPs, the detection performance of subnets is improved by increasing the number of parameters, which proves that by balancing the FLOPs and the number of parameters, not only can the requirement of lightweighting on oriented ship object detector for SAR images be satisfied, but also can effectively improve the detection performance of the lightweight detector.

3) Oriented Multiscale Attention: In the cross-attention layer of the decoder, in order to obtain accurate localization points of the ship features in SAR images, so as to extract effective ship features, we select different sampling points obtaining methods

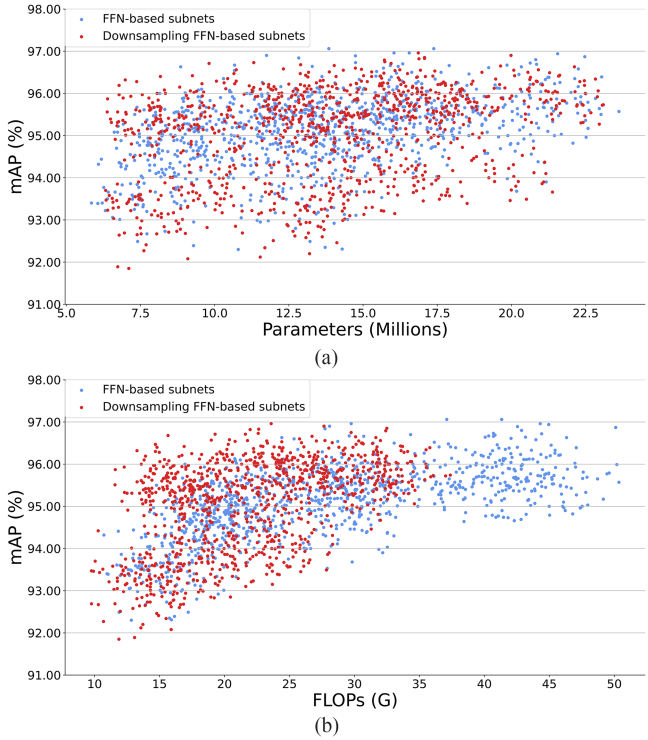


Fig. 8. (a) Comparison of parameters and detection performance before and after improvement of FFN. (b) Comparison of FLOPs and detection performance before and after improvement of FFN.

TABLE III
COMPARISON OF SAMPLING POINTS OBTAINING METHODS

Methods	Paras (M)	FLOPs (G)	Average inference time(ms)	Inshore AP.5	Offshore AP.5	All AP.5
MSDA [24]	5.81	10.86	21.77	87.44	98.54	95.43
O-MSDA	5.81	10.86	**21.40**	88.30	98.79	95.91
O-MSA (ours)	**5.78**	**10.84**	21.92	**89.30**	**99.23**	**96.68**

TABLE IV
COMPARISON OF DIFFERENT ENCODER AUXILIARY HEADS AND DECODER AUXILIARY TRAINING METHODS

Oriented R-CNN [32]	FPDDet [33]	Consistent encoding	Filter samples	All AP.5
√				95.33
	√			96.17
√	√			95.90
	√	√		96.32
	√	√	√	**96.68**

to compare the impact on the detection performance of long and slender ship objects, as shown in Fig. 9, which include MSDA, improved oriented multiscale deformable attention (O-MSDA), and our oriented multiscale attention (O-MSA). As shown in Table III, due to the fact that O-MSA does not learn the sampling offset, the number of parameters and FLOPs are lower compared

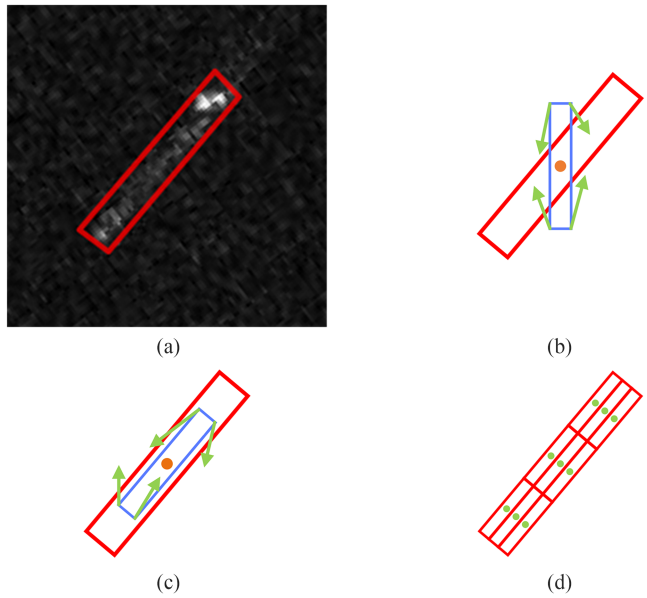


Fig. 9. (a) Input SAR ship image. (b) HBB-based sampling point acquisition strategy in multiscale deformable attention. (c) Our improved OBB-based sample point acquisition strategy in multiscale deformable attention. (d) Our proposed sample point acquisition strategy in oriented multiscale attention.

TABLE V
ABLATION EXPERIMENTS ON DIFFERENT DESIGNS BASED ON OUR TINY MODEL IN RSSDD

Auxiliary training	Downsampling FFN	O-MSA	One-to-many assignment	All AP.5	FLOPs (G)
				92.84	12.61
√				93.63	12.61
√	√			93.39	10.86
√	√	√		94.30	10.84
√	√	√	√	**96.68**	**10.84**

to the other two sampling point methods, while the experimental results of the comparison on our tiny model show that O-MSA is able to effectively improve the detection performance of the long and slender ship objects on the offshore scenario. Compared to MSDA offshore scenario improves 0.69%, and 1.25% AP gain is obtained in all scenarios, compared to O-MSDA offshore scenario improves 0.44%, and 0.77% AP gain is obtained in all scenarios, which proves that O-MSA not only reduces the number of parameters of the network, decreases the flops, and further lightweight the oriented ship object detector, but also reduces the background sampling of long and slender ship objects to improve the detection performance of oriented ship objects in SAR images.

4) Encoder Auxiliary Head and Decoder Auxiliary Training Method: In order to improve the encoder potential ship feature learning and the decoder cross-attention learning, we introduce an auxiliary head and decoder auxiliary training method to achieve the above goals. To apply the auxiliary head to the

TABLE VI
COMPARISON RESULTS OF DIFFERENT MODELS ON RSSDD

Models	Parameters (M)	FLOPs (G)	Average inference time (ms)	Resolution	Inshore AP.5	Offshore AP.5	All AP.5
DARTS- tiny [28]	13.21	26.99	11.29	608^2	35.01	72.07	60.09
DARTS- small [28]	26.99	44.83	13.83		39.70	73.23	62.62
DARTS- base [28]	46.14	69.02	14.73		36.44	76.26	63.52
Autoformer- tiny [31]	6.15	35.44	15.80		75.01	96.90	90.77
Autoformer- small [31]	14.49	83.54	20.88		76.02	98.34	92.18
Autoformer- base [31]	20.14	116.13	22.69		78.63	99.19	93.78
MobileNetV2 [58]	**5.51**	16.55	22.03	608^2	51.51	88.43	77.22
Rotated retinanet-R18 [34]	19.63	55.71	25.49		71.34	97.30	90.36
S2aNet-R18 [40]	22.04	50.99	39.04		84.46	98.14	94.57
Oriented R-CNN-R18 [32]	28.12	78.38	24.31		88.93	98.63	95.70
Ours- tiny	5.78	**10.84**	**21.92**		**89.30**	**99.23**	**96.68**
Rotated retinanet-R50 [34]	36.13	73.99	29.94	608^2	74.75	98.61	92.38
S2aNet-R50 [40]	38.54	69.27	43.48		87.23	98.77	95.46
SASM repppoints-R50 [59]	36.60	68.54	28.98		73.75	99.04	92.15
Oriented reppoints-R50 [60]	36.60	68.54	28.90		87.57	98.75	95.53
ReDet-ReR50 [61]	32.65	-	175.43		88.57	99.09	95.98
Quad-FPN-R50[62]	49.39	321.82	58.36		90.48	98.06	95.88
Oriented R-CNN-R50 [32]	41.13	97.44	**28.16**		89.79	**99.63**	96.62
Ours- small	**13.25**	**16.64**	29.26		**92.64**	99.48	**97.52**
Rotated retinanet-R101 [34]	55.12	101.40	38.02	608^2	76.25	98.33	92.55
S2aNet-R101 [40]	57.53	96.73	54.94		87.86	99.35	96.31
Sasm repppoints-R101 [59]	55.60	96.01	37.59		78.30	99.08	93.20
Oriented reppoints-R101 [60]	55.60	96.01	36.90		87.13	98.84	95.70
Quad-FPN-R101[62]	68.43	349.28	71.49		89.75	98.86	96.04
Oriented R-CNN-R101 [32]	60.12	124.91	36.10		91.72	99.40	97.02
Ours-base	**20.82**	**32.67**	**30.25**		**95.04**	**99.44**	**98.38**

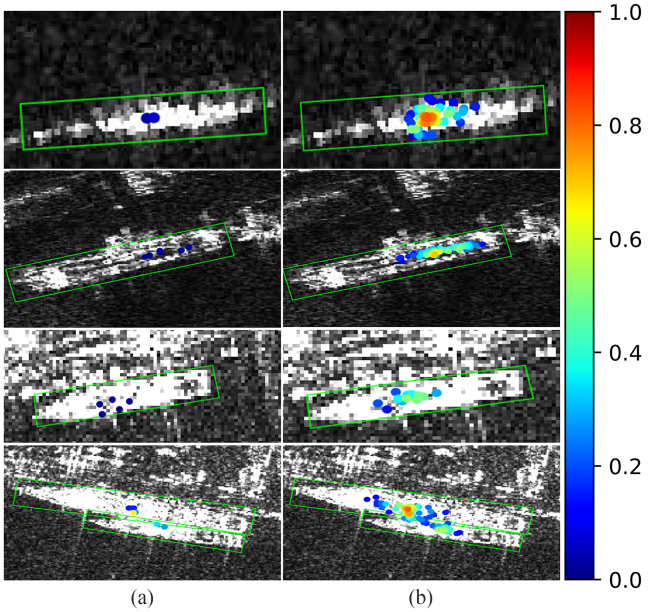


Fig. 10. (a) Visualizations of queries for the one-to-one Hungarian matching strategy. (b) Visualizations of queries for the one-to-many positive sample assignment strategy.

oriented ship object detection task, we compare the effects of different oriented object detection heads on the ship detection performance, as shown in Table IV. When FPDDet is used as the encoder auxiliary head, the detection performance in all scenarios is improved by 0.84% compared to the oriented R-CNN, and when two auxiliary heads are used, the AP value in all scenarios is reduced by 0.27%, which shows that the FPDDet auxiliary head is more suitable for learning the potential ship features in SAR images and improving the decoder cross-attention learning than other oriented object detection heads. Meanwhile, we further improved the OBB regression method of the FPDDet auxiliary head and the sample selection method used for auxiliary training of our decoder. As shown in Table IV, by adopting the OBB regression method consistent with the encoder output, the AP value of all scenarios gains 0.15%, which avoids the interference caused by the inconsistency of the OBB regression method between the auxiliary head and the encoder output, and considering that the FPDDet head uses a dense regression strategy to select positive samples, to further filter out the interference of low quality OBBs on the prediction results, we select the samples used for the auxiliary training of the decoder from the positive samples of the encoder according to the IoU threshold, which results in an 0.36% gain in the AP value of all scenarios, proving that the detection performance of the detector on ship objects could be improved by filtering out the low quality OBBs.

5) Ablation Experiments: Table V presents the ablation experiments of our proposed method based on our tiny model, which includes auxiliary training methods for the encoder and decoder, downsampling FFN for the encoder, O-MSA in the decoder, and one-to-many positive sample assignment strategy for the outputs of the encoder and decoder. By using auxiliary training to improve the learning of potential ship features in the encoder and the learning of cross-attention in the decoder, an AP gain of 0.79% is obtained, and at the same time, the auxiliary training part can be discarded in the inference phase, so it does not increase the FLOPs of our model. The downsampling FFN slightly reduces the detection performance, but the network FLOPs are significantly reduced, by increasing the number of

TABLE VII
COMPARISON RESULTS OF DIFFERENT MODELS ON RSDD

Models	Parameters (M)	FLOPs (G)	Average inference time (ms)	Resolution	Inshore AP.5	Offshore AP.5	All AP.5
Autoformer-tiny [31]	6.64	27.15	12.13	512^2	65.61	94.06	89.07
Autoformer-small [31]	13.22	54.08	13.49		69.08	94.93	90.21
Autoformer-base [31]	20.34	83.19	17.67		70.46	95.07	90.82
MobileNetV2 [58]	7.33	17.37	**16.38**	512^2	30.28	80.79	71.31
S2aNet-R18 [40]	22.04	36.11	34.68		59.61	92.50	86.88
Rotated retinanet-R18 [34]	19.63	39.46	20.45		30.10	72.74	65.09
Oriented R-CNN-R18 [32]	28.12	63.67	20.43		61.82	90.05	87.50
Ours-tiny	**6.37**	**8.85**	19.90		**71.67**	**96.32**	**92.19**
S2aNet-R50 [40]	38.54	49.05	40.48	512^2	63.27	93.14	87.91
Rotated retinanet-R50 [34]	36.13	52.39	24.09		33.20	74.06	66.66
ReDet-ReR50 [61]	32.65	-	172.41		61.94	90.34	88.40
Quad-FPN-R50[62]	49.39	236.30	46.81		68.77	95.14	90.82
Oriented R-CNN-R50 [32]	41.13	77.18	23.25		65.92	90.21	88.84
Ours-small	**12.94**	**13.91**	**23.14**		**72.40**	**96.54**	**92.57**
S2aNet-R101 [40]	57.53	68.53	52.08	512^2	66.43	94.94	90.06
Rotated retinanet-R101 [34]	55.12	71.87	28.49		35.75	74.92	67.89
R3Det-R101 [18]	60.57	101.65	33.22		57.73	90.09	80.92
Polar Encoding-R101 [20]	74.37	180.19	47.91		62.02	89.99	87.88
Quad-FPN-R101[62]	68.43	255.78	54.61		70.43	95.37	91.17
Oriented R-CNN-R101 [32]	60.12	96.66	28.08		66.77	90.28	88.85
Ours-base	**25.78**	**32.14**	**27.63**		**78.36**	**96.57**	**93.45**

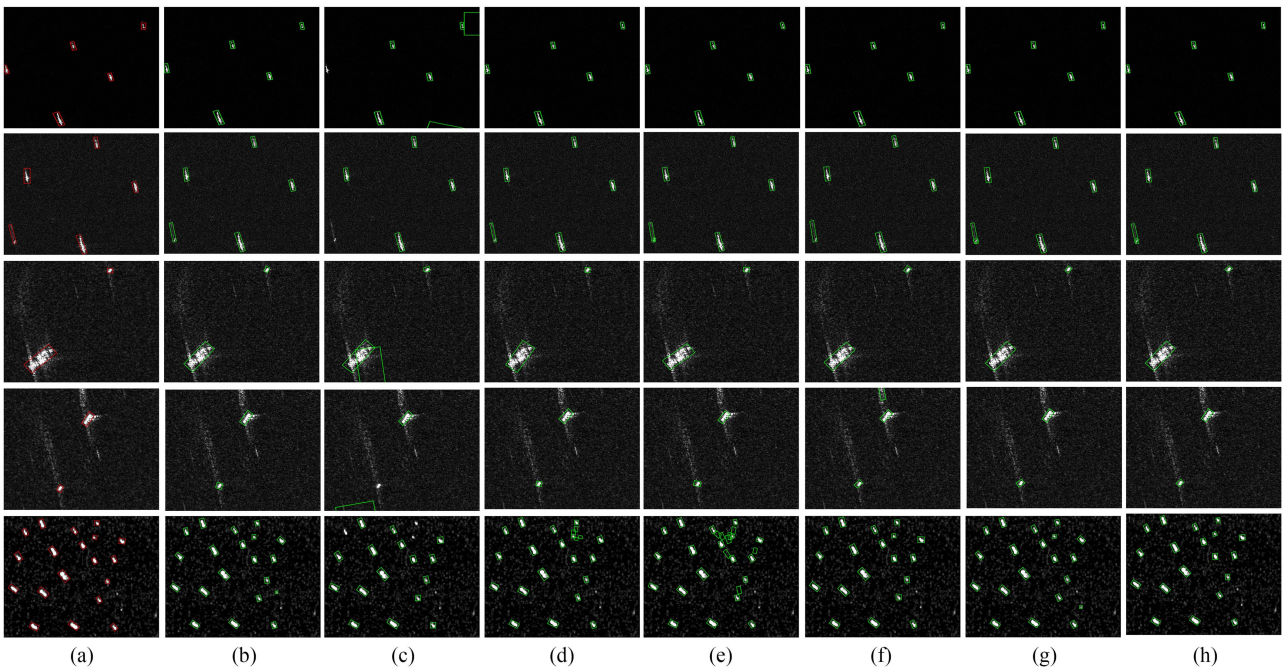


Fig. 11. Visualization results in the RSSDD offshore scenario. (a) Ground truth. (b) Our base model. (c) MobileNetV2. (d) Rotated retinanet-R101. (e) S2aNet-R101. (f) SASM repppoints-R101. (g) Oriented reppoints-R101. (h) Oriented R-CNN-R101.

network parameters can effectively improve the detection performance of SAR image ships while satisfying the lightweight requirement of the network. O-MSA not only further reduces the number of network parameters and FLOPs by eliminating sampling offset learning compared to MSDA [24], but also achieves 0.91% AP gain. As shown in the visualization of the decoder queries in Fig. 10(a) and (b), the one-to-many positive sample assignment strategy has more high-quality prediction results and higher prediction scores than the one-to-one assignment strategy. Considering the interference of the complex background of the SAR image on the ship objects and the interference of the negative samples of the ship objects on the positive samples, the one-to-many positive sample assignment strategy can effectively alleviate the above interferences and

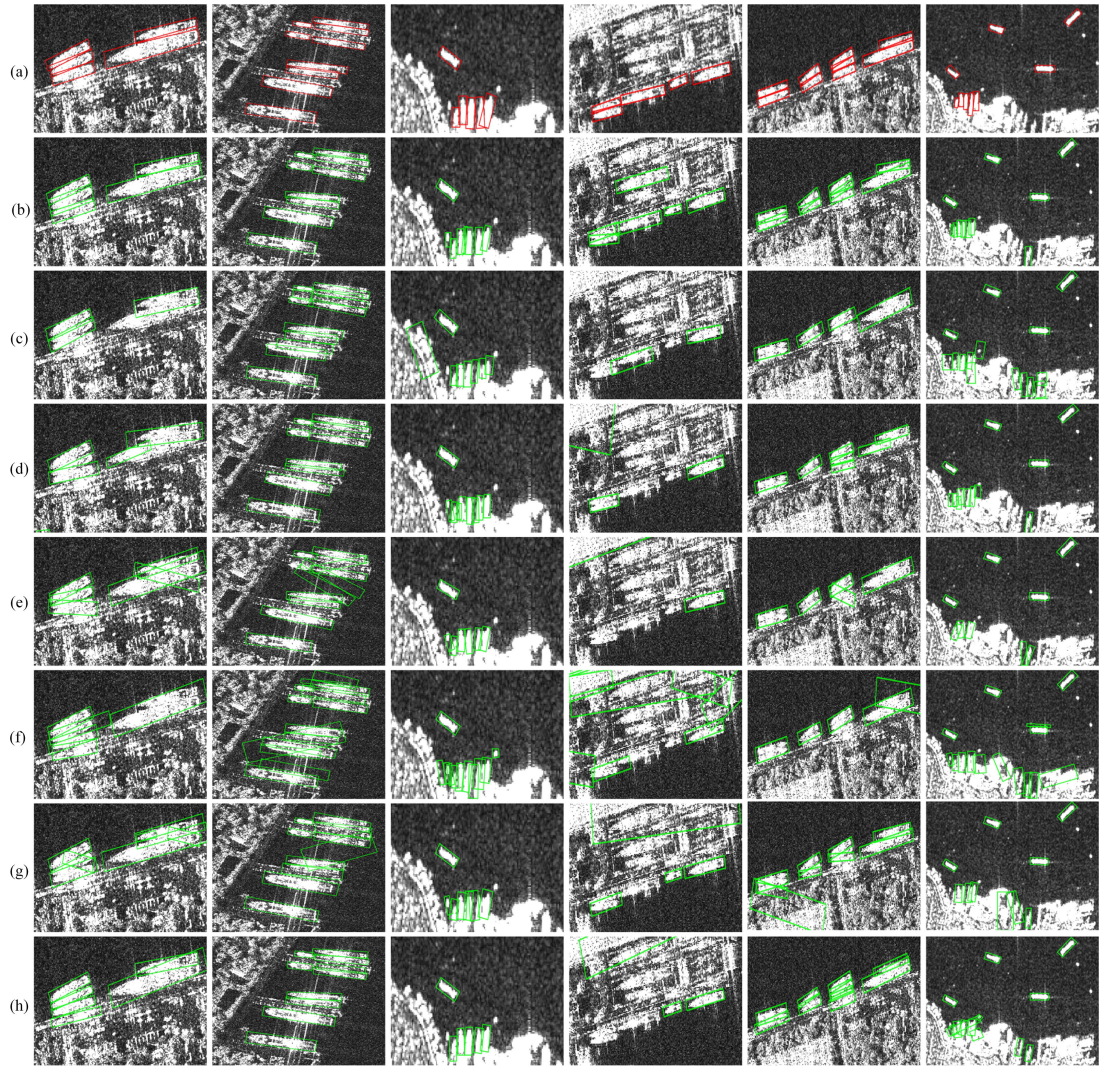


Fig. 12. Visualization results in the RSSDD inshore scenario. (a) Ground truth. (b) Our base model. (c) MobileNetV2. (d) Rotated retinanet-R101. (e) S2aNet-R101. (f) SASM repppoints-R101. (g) Oriented reppoints-R101. (h) Oriented R-CNN-R101.

achieve 2.38% AP gain. In conclusion, the ablation experiments in Table V have demonstrated that each of the designs in our model is effective in lightweighting the oriented SAR ship object detector and improving ship detection performance, and that our model finds a balance between lightweighting and detection performance.

C. Comparison With the Oriented Object Detection Methods

To validate the advantages of our methods in terms of lightweighting and detection performance of oriented SAR ship object detector. As shown in Table VI, the comparison results with the NAS-based models and mainstream oriented object detectors are presented, our tiny model reduces the FLOPs by 105.29G, the average inference time by 0.77 ms, and the detection performance of all scenarios achieves 2.90% AP gain compared to the Autoformer-base model in NAS-based method, and compared to the manual design based MobileNetV2 model, the FLOPs are reduced by 5.71G, the average inference time is reduced by 0.11 ms, and the detection performance of all scenarios achieves 19.46% AP gain. Our small model also achieves optimal performance compared to other models while reducing network FLOPs, with 0.37% AP gain for inshore scenarios, 0.60% for offshore scenarios, and 0.98% gain in AP values for all scenarios. Our base model reduces the number of parameters by 65.37%, the FLOPs by 92.24G, and the average inference time by 5.85 ms, while achieving an 1.36% gain in AP value for all scenarios compared to the previous state-of-the-art model, oriented R-CNN. Comparison experiments show that our model not only achieves a lightweight network with lower network parameters, FLOPs, and average inference time compared to previous models, but also achieves state-of-the-art performance, achieving 98.38% AP on RSSDD with only 20.82M number of parameters, 32.67G FLOPs and 30.25 ms average inference time.

To further qualitatively verify the superiority of our models in terms of detection performance, Figs. 11 and 12 show the visualization results of different detectors on RSSDD for offshore

and inshore, respectively. The offshore detection visualization results in Fig. 11 can further validate the accuracy of our models for thin and slender ship object detection. The inshore detection visualization results in Fig. 12 show that our model is able to accurately detect densely arranged ship objects in complex backgrounds, and further demonstrate that the one-to-many positive sample assignment strategy can effectively alleviate the interference of complex backgrounds and negative samples of ship objects on inshore ship detection.

D. Experimental Results on RSDD

Table VII shows the comparison results of our model with various models on RSDD, our tiny model improves the AP value of all scenarios by 1.37%, and reduces the FLOPs by 74.34G compared to the NAS-based Autoformer-base model, and improves the AP by 4.69%, reduces the number of parameters by 21.75M, and reduces the FLOPs by 54.82G compared to the ResNet-18-based models. Compared to the previous models with ResNet-50 as the backbone, our small model reduces the number of parameters by 36.45M, reduces the FLOPs by 222.39G, and improves the AP value of all scenarios by 1.75%. Our base model not only achieves the lightweighting of the oriented object detector for SAR images, but also achieves the 2.28% improvement in AP value and the state-of-the-art performance in RSDD when compared to various models with ResNet-101 as the backbone. The comparison results prove that our model can be applied to a wide range of SAR image ship detection scenarios and achieve excellent detection performance.

V. Conclusion

This article proposes an NAS-based lightweight detector in polar coordinate system with DETR as the search space for oriented SAR ship detection. By encoding and decoding OBB in polar coordinate system, the HBB-based DETR models are applied to the task of oriented SAR ship detection and effectively avoid the boundary discontinuity problem brought by mainstream OBB encoding methods, and in order to realize the lightweight and automatic design of the DETR model, we build the detector based on the weight entanglement strategy in NAS. At the same time, we introduce downsampling FFN in the encoder to effectively reduce the overall FLOPs of the network. In addition, we design an oriented multiscale attention in the decoder to alleviate the problem of sampling a large amount of background when performing offset learning for thin and slender ship objects and improve the detection performance of offshore ship objects. Then, the FPDDet head is introduced to improve the learning of potential ship features in the encoder and cross-attention learning in the decoder. Finally, a one-to-many positive sample assignment strategy is used to alleviate the interference of complicated backgrounds and the negative samples of ship objects on positive samples. Our model not only has fewer parameters, lower FLOPs, and shorter average inference time, but also has superior detection performance compared to previous models on RSSDD and RSDD, while our base model achieves state-of-the-art performance, achieving 98.38% AP on RSSDD with 20.82M number of parameters, 32.67G FLOPs and 30.25 ms average inference time, and 93.45% AP on RSDD with 25.78M number of parameters, 32.14G FLOPs and 27.63 ms average inference time.

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